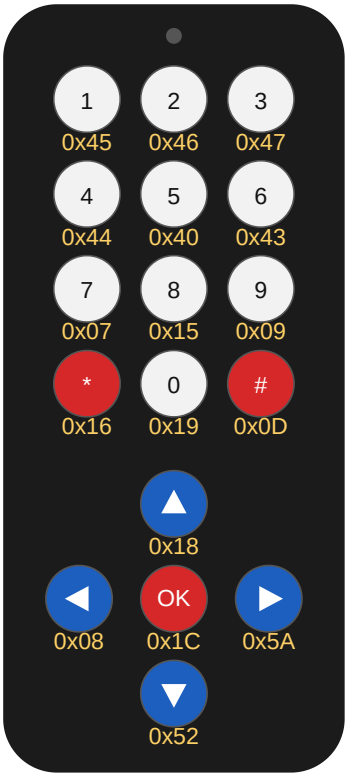
NEC IR remote control HX1838 – code table

17-key remote from Arduino kits · address 0x00 · 38 kHz carrier



Command code (hex) below each key

Frame format

Leader	9 ms burst + 4.5 ms space
Byte 1	Address = **0x00**
Byte 2	~Address = **0xFF**
Byte 3	Command (see table)
Byte 4	~Command (check)
Stop	562.5 µs burst

Each byte is sent **LSB first**. Byte 3 + byte 4 = 0xFF in a valid frame.

Bit encoding

Logic 0	562.5 µs burst + 562.5 µs space = 1.125 ms
Logic 1	562.5 µs burst + 1687.5 µs space = 2.25 ms
Burst	≈ 21 periods of 38 kHz (26.3 µs)

Repeat code

While the key is held, the remote only sends
9 ms burst + 2.25 ms space + 562.5 µs burst
about every 108 ms (measured from the start of the frame).

Receiver (HX1838 / TSOP)

The output is **active low**: low during a burst, high during space and idle.
Decode by measuring the **high time** after each low pulse.

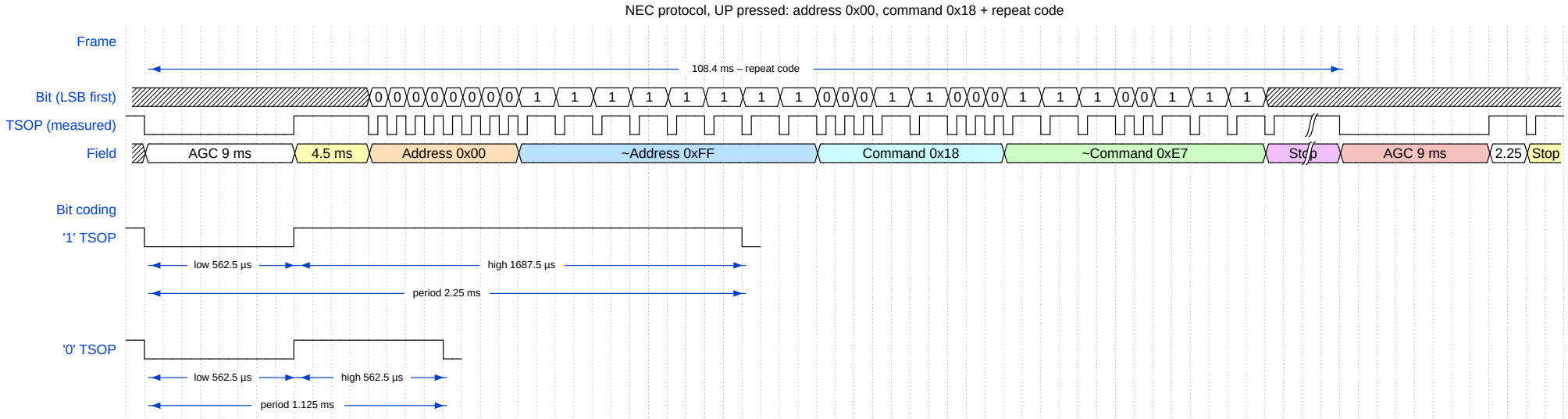
Code table

Key	Command	~Command	Command bits LSB first, as transmitted
1	0x45	0xBA	10100010
2	0x46	0xB9	01100010
3	0x47	0xB8	11100010
4	0x44	0xBB	00100010
5	0x40	0xBF	00000010
6	0x43	0xBC	11000010
7	0x07	0xF8	11100000
8	0x15	0xEA	10101000
9	0x09	0xF6	10010000
*	0x16	0xE9	01101000
0	0x19	0xE6	10011000
#	0x0D	0xF2	10110000
UP	0x18	0xE7	00011000
LEFT	0x08	0xF7	00010000
OK	0x1C	0xE3	00111000
RIGHT	0x5A	0xA5	01011010
DOWN	0x52	0xAD	01001010

UP (0x18) is verified against a measured signal (page 2). The other codes are the standard values for this type of remote – clones may differ, so check with your own receiver.

Measured signal: the UP key

nec_Bt_up_rec.csv · TSOP output sampled every 25 µs · address 0x00, command 0x18 + repeat code



Frame: 1 tick = 1.125 ms · Bit coding: to scale, 562.5 µs = 8 ticks · TSOP output is active low · gap (|) = 39 ms without signal, shortened

How to decode the signal

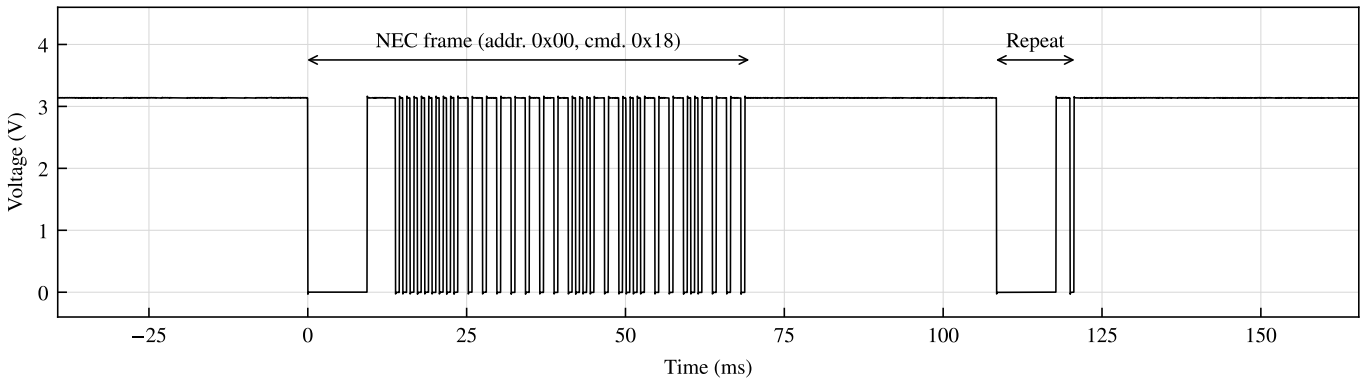
1. Wait for a falling edge (start of a burst).
2. Measure the low time ≈ 9 ms, then the high time: ≈ 4.5 ms = new frame, ≈ 2.25 ms = repeat code.
3. For each of the 32 bits: measure the high time after the short low pulse. ≈ 0.56 ms → 0, ≈ 1.69 ms → 1. Use a threshold of e.g. 1.1 ms.
4. Assemble the bytes with the first bit as bit 0 (LSB first).
5. Check that command + ~command = 0xFF.

Measured vs. nominal timing

	Nominal	Measured
Leader low	9.0 ms	9.35 ms
Leader high	4.5 ms	4.48 ms
Bit low (burst)	562.5 µs	600–625 µs
Bit high, 0	562.5 µs	525–550 µs
Bit high, 1	1687.5 µs	1625–1650 µs
Frame → repeat	≈ 108 ms	108.4 ms

The TSOP receiver stretches the low pulses slightly and shortens the spaces by the same amount. A decoder must therefore use tolerances, not exact times.

Oscilloscope measurement



Measured output of the TSOP IR receiver (active low) for the UP key: NEC frame with address 0x00 and command 0x18, followed by a repeat code 108.4 ms after the start of the frame.